PHYSIOLOGY

Comparative exploratory behavior of *Acomys cahirinus* and *Mus musculus*

Natalia Merkulyeva[1], Aleksandr Veshchitskii[1], and Evgeniya Efimova[2]

[1] Pavlov Institute of Physiology of the Russian Academy of Sciences, nab. Makarova, 6, St. Petersburg, 199034, Russian Federation
[2] St. Petersburg State University, Universitetskaya nab., 7–9, St. Petersburg, 199034, Russian Federation

Address correspondence and requests for materials to Merkulyeva Natalia, merkulyevan@infran.ru

Abstract

Acomys cahirinus is a promising model in neuroscience. Their hippocampus is more developed than that of mice. Given the strong relationship between the hippocampus and exploratory behavior, we conducted a comparative investigation of behavior in acomyses and mice using open field, elevated plus-maze, and light/dark box tests. Compared to mice, acomyses: (1) exhibited lower moving activity in the open field test, characterized by reduced moving time, distance walked, and walking velocity; (2) spent more time in the open arms and less time in the central platform of the elevated plus-maze, while spending similar time in the central part of the open field arena; (3) displayed more rearing behavior in the open field test; (4) perform shorter latencies to enter the light compartment in the light/dark box. Significant differences may be attributed to the different exploratory behavior and non-equal levels of the anxiety of these species.

Keywords: Exploratory behavior, *Acomys cahirinus, Mus musculus*, Open field, Elevated plus-maze, Light/dark box.

Introduction

Behavioral tests are widely used in various investigations related to general physiology, neurobiology, and pharmacology (see: Henry et al., 2010), particularly in studies of neuronal development (Rampon and Tsien, 2000) and central nervous system repair (Balkaya et al., 2018; Callaghan, Yang, Moloney, and Waeber, 2025). Exploratory behavior in unfamiliar environments is one of the most easily observed and controlled forms of behavior in captivity (Birke and Sadler, 1986). Common tests used to study exploratory behavior include the open field test, the elevated plus-maze, and the light/dark box test (Prut and Belzung, 2003; Lever and Burton, 2006; Gould, Dao, and Kovacsics, 2009; Ratnayake et al., 2012; Sturman, Germain, and Bohacek, 2018). The open field test is commonly used to assess exploratory behavior, general locomotor activity, and anxiety (see: Gould, Dao, and Kovacsics, 2009). The elevated plus-maze exploits rodents' innate conflict between exploration and avoidance of potentially dangerous areas (Cryan and Sweeney, 2011). The elevated plus-maze and light/dark box tests also take advantage of the innate aversion of rodents to brightly illuminated areas (Crawley and Goodwin, 1980).

All laboratory animals should undergo appropriate behavioral testing to establish a baseline for their future use in experimental studies. Over time, new animal models have inevitably been developed. One such model is represented by animals from the *Acomys* genus (*A. cahirinus*, *A. dimidiatus, A. russatus, A. kempi, A. percivali,* and others) within the order *Rodentia*. These species exhibit unique regenerative capabilities being precocial animals with highly developed regen-

Citation: Merkulyeva, N., Veshchitskii, A., and Efimova, E. 2025. Comparative exploratory behavior of *Acomys cahirinus* and *Mus musculus*. Bio. Comm. 70(4): 276–283. EDN EOANSZ

Authors' information: Natalia Merkulyeva, PhD in Biology, Laboratory Head, Leading Researcher, orcid.org/0000-0003-1276-1918; Aleksandr Veshchitskii, PhD in Biology, Senior Researcher, orcid.org/0000-0002-9720-8902; Evgeniya Efimova, PhD in Biology, Leading Researcher, orcid.org/0000-0002-9641-515X

Manuscript Editor: Ludmila Stankevich, Department of Physiology of Higher Nervous Activity and Psychophysiology, St. Petersburg State University, St. Petersburg, Russia

Received: July 30, 2025;

Revised: October 4, 2025;

Accepted: October 6, 2025.

Funding: This study was supported by the state funding allocated to the Pavlov Institute of Physiology Russian Academy of Sciences (no. 1021062411653-4-3.1.8).

Ethics statement: This paper does not contain any studies involving human participants or animals performed by any of the authors.

Competing interests: The authors have declared that no competing interests exist.

erative abilities (see: Merkulyeva, 2024; Tomasso, Disela, Longaker, and Bartscherer, 2024).

Despite limited knowledge regarding the structure of the acomys brain, some data suggest that the hippocampus of acomys is more developed than that of mice and rats (D'Udine and Gozzo, 1983; Dmitrieva, Danilenkova, Gozzo, and D'Udine, 1987). Given the strong relationship between exploratory behavior and the hippocampus (Lever, Burton, and O'Keefe, 2006; Crusio, Schwegler, Brust, and Van Abeelen, 1989; Johnson et al., 2012; Zeng et al., 2024), a comparative investigation of exploratory behavior in acomyses appears particularly valuable. Several studies addressed exploratory behavior of acomys — strategies they use to explore (Birke and Sandler, 1986), dependence of their locomotor activity on the lighting and environment (Eilam, 2004), personal traits in their behavior (Frynta et al., 2024). Also, a series of studies were performed comparing exploratory behavior of acomys and *Mus musculus*. In those, an exploration of the novel arena, as well as exploration of novel objects, tendency to use burrows and reaction to "predator" were assessed (Birke, D'Udine, and Albonetti, 1985; D'Udine, Dimitriev, and Birke, 1987). Acomys showed to be more active in their exploration of novel areas than mice, spending less time to leave the familiar area and also spending less time in provided shelter (Birke, D'Udine, and Albonetti, 1985; D'Udine, Dimitriev, and Birke, 1987). It should be noted that in those papers the arena was not completely empty; instead, an arena was subdivided into several sectors or had a cross-like structure in the center. The modification of the arena, as well as conditions of the experiment can *per se* be separate quests for animals and influence their behavior.

Modern standardized tests, such as simple open field described above, elevated plus-maze and light dark box are commonly used to assess the effects of various experimental interventions on the general behavior in the testing environment in *Mus musculus*. However, the behavior in such tests have not yet been studied for acomys in comparison with *Mus musculus*. Therefore, the aim of the present study was to conduct a comparative analysis of exploratory behavior in *A. cahirinus* (acomys) and the mouse.

Material and methods

Animals and care

Six adult acomyses (*Acomys cahirinus*) and six adult C57BL/6/129SvEv mice (*Mus musculus*) of both sexes (3 males and 3 females), aged 3 months, were used in the study. The acomyses were bred in the neuromorphology lab of the Pavlov Institute of Physiology RAS. All animals were housed in a temperature-controlled room (24–25 °C) with 50 % humidity, maintained under a 12-hour light/dark cycle. Each propylene cage, measuring 530 × 325 mm, accommodated 3–5 individuals. Water was provided ad libitum, and food was supplied according to a diet formulated in our laboratory (Shkorbatova et al., 2024). The mice were bred in the vivarium of St. Petersburg State University. Note that different husbandry conditions are a consequence of testing different species with varying ecological requirements. Previous studies have shown no strong gender dependencies in behavior in the open field (Sturman, Germain, and Bohacek, 2018), elevated plus maze (Tucker and McCabe, 2017), and light/dark box (Tsao et al., 2023) in mice. Similar findings were reported for acomyses in the open field and elevated plus-maze (Ratnayake et al., 2014). Consequently, mixed-gender samples were used in the present study.

All experiments were conducted in compliance with the Council Directive 2010/63EU of the European Parliament regarding the protection of animals used for scientific purposes and were approved by the Ethics Commission of the Pavlov Institute of Physiology (Protocol no. 03/10; 10.03.2025) and by the Ethics Commission of the St. Petersburg State University (Protocol no. 131-03-3; 13.03.2024).

Experimental design

Three tests were used to assess exploratory behavior in novel environments: the open field test, the elevated plus-maze, and the light/dark box test. Video recordings were captured using a camera placed above the center of the testing area. Before testing, each animal was removed from the group cage and placed individually in a single cage (identical in size and shape to the home cage) for 15 min. After each test, the animal was returned to its cage, and all testing platforms were cleaned with 10 % ethanol. Following a 5-minute interval, the next animal was placed in the test. Post-acquisition analysis of the activity was conducted using Noldus Ethovision software to track the movement of the animals. A gap lasting for 30 min was used between different tests.

Behavioral tests

Both mice and acomyses are nocturnal species (Cohen et al., 2010; Morrow, Smale, Meek, and Lundrigan, 2024), therefore, all animals were tested after 12 PM when their activity was elevated.

Open field test. A large round open area made of grey polyvinyl chloride 63 cm in diameter with 50 cm high walls was used as an open field. The illumination level was set at 10 Lx. Animals were individually placed in the open area for a duration of 10 min, as previously proposed (Sturman, Germain, and Bohacek, 2018; Prut and Belzung, 2003). The following behavioral patterns were recorded: time spent in the central and peripheral

PHYSIOLOGY

regions of the arena, walking distance, walking velocity, total moving and non-moving time, and number of rearing.

Elevated plus-maze test. The elevated plus-maze was constructed from polyvinyl chloride and consisted of two open arms (35 cm long and 5 cm wide) and two closed arms (of the same dimensions, with 15 cm high walls) elevated 50 cm above the floor. The illumination level was set at 10 Lx. Animals were individually placed in the maze for a duration of 5 min, as previously proposed (Rico et al., 2017; Tucker and McCabe, 2017). Following behavioral parameters were recorded: walking distance and time spent in the closed, open, and central areas of the maze.

Light/dark box test. The light/dark box was constructed from polyvinyl chloride and consisted of two chambers: a dark chamber with a roof (20 cm long and 20 cm wide) and a light chamber without a roof (also 20 cm long and 20 cm wide) with walls 20 cm. The illumination level in the light chamber was set at 110 lux. Animals were individually placed in the box in a light chamber and recorded for 3 min, as previously proposed (Shimada et al., 1995; Nöldner and Schötz, 2007). Post-acquisition analysis of the activity was conducted manually, assessing the total time spent in the light chamber, latency to enter dark chamber, latency to transition from dark chamber into light chamber, as well as the total number of transitions between the light and dark chambers.

Statistical analysis

Data is presented as median (Q1, Q3). Data were analyzed using R (RStudio Version 2024.09.1+394). First, the distribution of each variable was assessed using the Shapiro — Wilk test to evaluate normality. Since some variables were abnormally distributed, a non-parametric Mann — Whitney t-test was applied.

Results

Open field

One of the striking differences between acomyses and mice was the balance between moving and non-moving activity within the open field arena. Acomyses exhibited significantly less moving time compared to mice (388 ± 24 s vs 484 ± 61 s; Mann — Whitney test: $p = 0.026$) (Fig. 1A). The total walking distance in acomyses was approximately half of that of mice

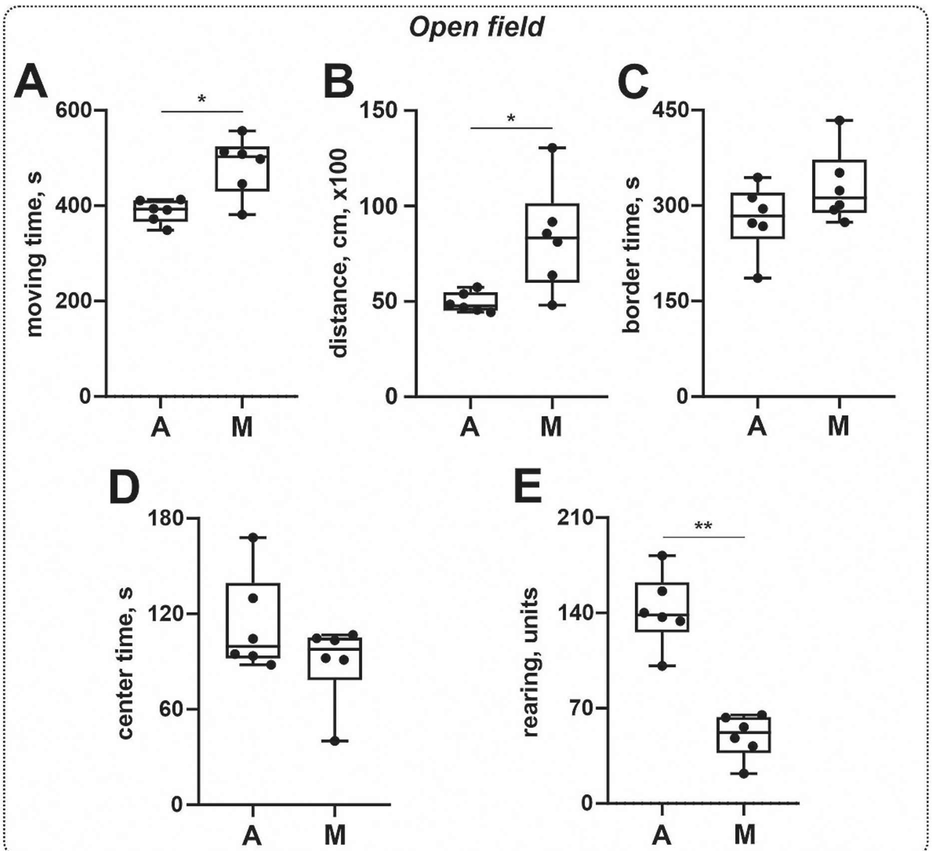

Fig. 1. Analysed behavioral parameters in the open field test for acomyses (A) and mice (M): A — moving time; B — distance walked in 10 min; C — time spent in the border area; D — time spent in the central area; E — rearing frequency. * $p < 0.05$; ** $p < 0.01$.

(4943 ± 519 cm vs 8344 ± 2804 cm; Mann — Whitney test: p = 0.015) (Fig. 1B). Additionally, the walking velocity in acomyses was lower than in mice (8.28 ± 0.90 cm/s vs 13.96 ± 4.73 cm/s; Mann — Whitney test: p = 0.015).

When comparing time spent at the border *versus* the central part of the arena, no significant differences were observed between acomyses and mice (border time: 279.6 ± 53.6 s vs 329.5 ± 57.6 s; Mann — Whitney test: p = 0.180; central time: 113.0 ± 30.7 s vs 89.7 ± 25.17 s; Mann — Whitney test: p = 0.485) (Fig. 1C, D).

Acomyses displayed a significantly higher frequency of rearing (both with and without support) compared to mice (141.7 ± 26.7 vs 49.3 ± 16.0; Mann — Whitney test: p = 0.002) (Fig. 1E).

It is noteworthy that one mouse exhibited moving time, walking distance, and walking velocity values similar to those of acomyses. However, other parameters of open field activity for this mouse fell within the "mice" spectrum.

Elevated plus-maze

The total distance walked in the maze by acomyses was comparable to that of mice (1457 ± 214 cm vs 1391 ± 203 cm; Mann — Whitney test: p = 0.699) (Fig. 2A). However, acomyses spent significantly more time in open arms compared to mice (72.8 ± 20.3 s vs 39.9 ± 25.1 s; Mann — Whitney test: p = 0.026) (Fig. 2B). The time spent in closed arms was similar for both groups (129.4 ± 24.3 s vs 152.8 ± 20.2 s; Mann — Whitney test: p = 0.180) (Fig. 2C). In contrast, acomyses spent significantly less time at the central platform compared to mice (77.3 ± 14.4 s vs 107.3 ± 16.9 s; Mann — Whitney test: p = 0.026) (Fig. 2D). The most striking difference was a rearing behavior: acomyses performed significantly more rearings in the closed arms (34.2 ± 10.5 vs 15.3 ± 6.1; Mann — Whitney test: p = 0.0087) (Fig. 2E) and, in contrast to mice, performed rearings (ranged 2–4) in open arms (Fig. 2F).

Light/dark box

The total time spent in the light chamber by acomyses was comparable to that of mice (48.4 ± 12.1 s vs 58.1 ± 24.5 s; Mann — Whitney test: p = 0.589) (Fig. 3A). The number of transitions from the light to the dark chamber was similar between acomyses and mice (7.0 ± 2.4 unit/3 min vs 9.5 ± 3.8 unit/3 min; Mann — Whitney test: p = 0.418) (Fig. 3B). The latency period for the first transitioning from the light to the dark chamber was also comparable

PHYSIOLOGY

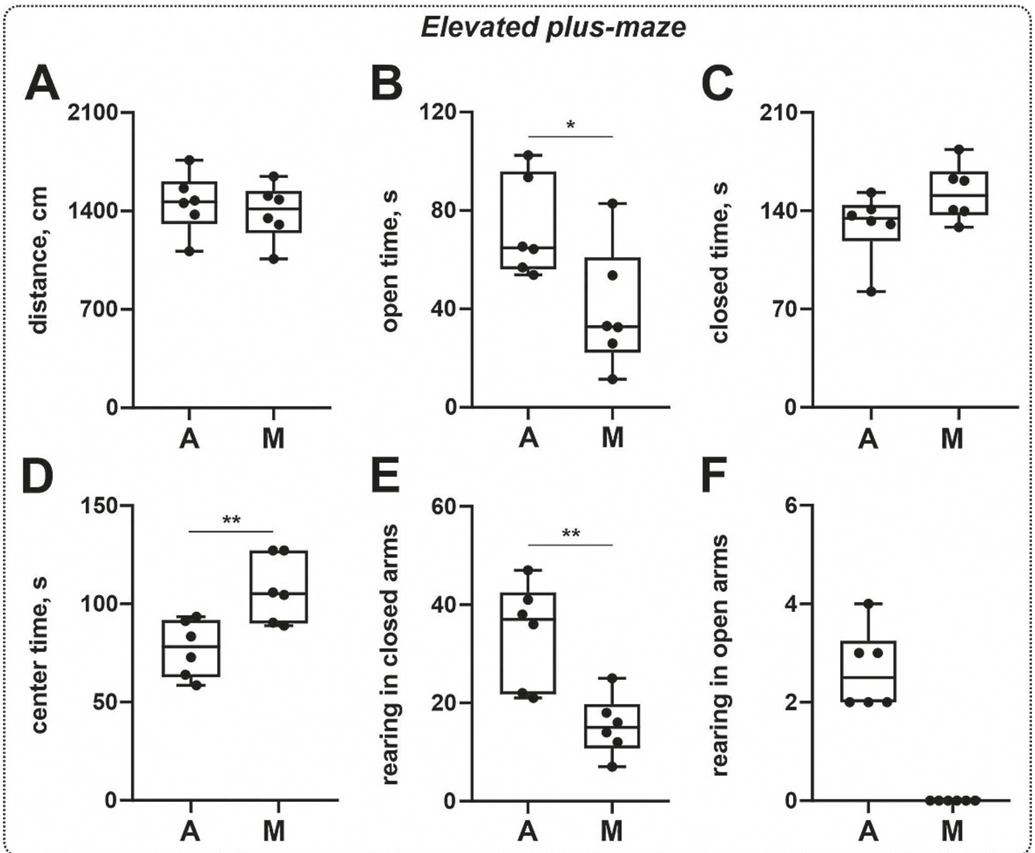

Fig. 2. Analyzed behavioral parameters in the elevated plus-maze test for acomyses (A) and mice (M): A — distance walked in 5 min; B — time spent in open arms; C — time spent in closed arms; D — time spent at the central platform; E — rearing frequency in closed arms; F — rearing frequency in open arms. * p < 0.05, ** p < 0.01.

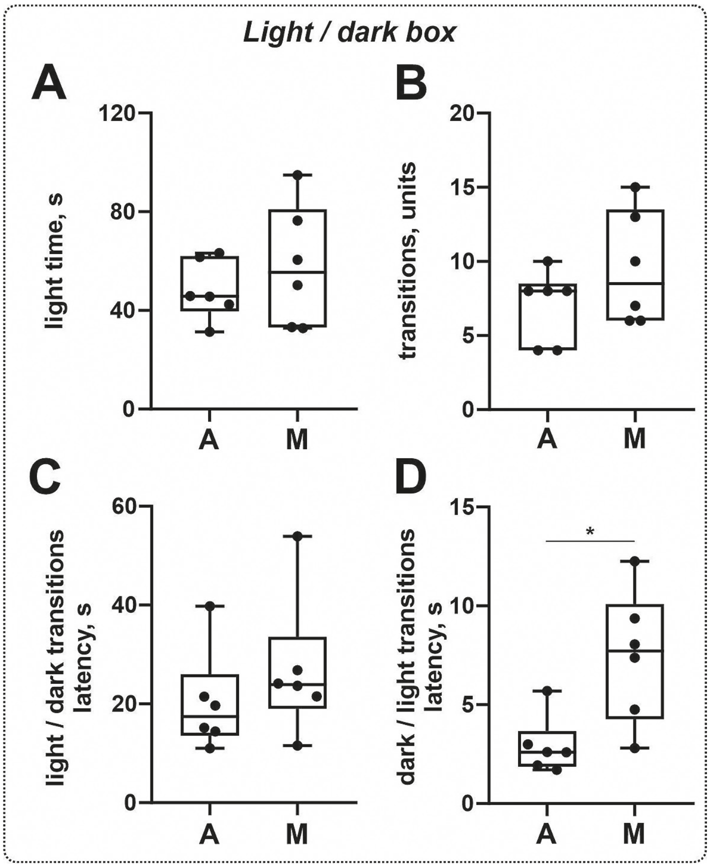

Fig. 3. Analyzed behavioral parameters in the light/dark box test for acomyses (A) and mice (M): A — time spent in the light; B — frequency of transitions; C — latency for light/dark transitions; D — latency for dark/light shift transitions. * p < 0.05.

in both groups (20.3 ± 10.3 s vs 26.9 ± 14.2 s; Mann — Whitney test: $p = 0.180$) (Fig. 3C). However, the latency period for transitioning from the dark to the light chamber (emergence latency) was significantly shorter in acomyses (2.9 ± 1.4 s vs 7.4 ± 3.3 s; Mann — Whitney test: $p = 0.015$) (Fig. 3D).

Discussion

The lower moving activity of acomyses in the open field test is accompanied by a higher rate of rearing and the less time to transit from the dark to the light cameras in the ligh/dark box test. This may reflect the higher exploratory activity of acomys. Moreover, specific behavioral patterns in the open field test were reported for different rodent species (Nauman, 1968; Hagbi and Eilam, 2022; Rogov et al., 2025). One structure tightly related to the integration of sensory stimuli and to the exploratory behavior, is hippocampus. In acomys, it is more developed with respect to the neocortex than in rats and mice (D'Udine and Gozzo, 1983; Dmitrieva et al., 1987). Based on these data, the distinctive orienting behaviors of the acomys was proposed (D'Udine and Gozzo, 1983; Gozzo et al., 1985). Differences in exploratory behavior in a novel environment between acomys and mouse (including different locomotor patterns) were also observed (Birke, D'Udine, and Albonetti, 1985; D'Udine, Dimitriev, and Birke, 1987) — both works suggested higher exploratory behavior in novel environments of acomys, as they spent less time in familiar starting box, had lower latency time to enter the novel environment and had explored more sections in open field arena compared to mice. In previous works, no differences in rearing frequency between acomyses and mice was reported (D'Udine, Dimitriev, and Birke, 1987; Birke, D'Udine, and Albonetti, 1985); it is contradictory with data obtained here and with the report about in acomys, the rearing is a highly repeatable behavioral pattern (Frynta et al., 2024). Interestingly, authors in previous works expected to see an increased number of rearings, due to work of Wilson et al. (1976) stating that in desert-dwelling species rearings are more common (Wilson, Vacek, Lanier, and Dewsbury, 1976). It is possible that differences we have seen can be due to different conditions used during the tests — in Birke et al.'s (1985) and D'Udine et al.'s (1987) studies, behavioral assessment was done under dim red light and during dark periods of time. Also, different strains of mice were used in the studies of (Birke, D'Udine, and Albonetti, 1985; D'Udine, Dimitriev, and Birke, 1987): not inbred C57BL/6/129SvEv mice, but other strains were used (outbred white Swiss mice and hybrid B6CBAF1 mice). At the same time, behavioral differences between different rodent strains were previously reported (Hall et al., 2000; Clemens et al., 2014; Keeley, Bye, Trow, and McDonald, 2015).

The differences in the exploratory behavior also can be related to the fact that acomyses are more relayed upon the visual cues compared to mice. It is important to note that although specific data on visual acuity in acomyses relatively to mice are lacking, evidence suggests that acomyses have better depth perception than gerbils (Greenberg, 1986), and that gerbils possess more developed vision than mice (Günter, Belhadj, Seeliger, and Mühlfriedel, 2024). These findings imply a greater development of vision in acomyses compared to mice.

On the other hand, the above-mentioned differences can be related to the lower level of anxiety in acomyses, especially in the light of the data about acomyses that spent more time on the open platform of the elevated plus-maze. Note that all three tests are also used for the anxiety-like behavior assessment (Prut and Belzung, 2003; Kraeuter, Guest, and Sarnyai, 2018; Kulesskaya and Voikar, 2014). Interesting that in contrast to mice, acomyses performed rearings in the open arms that possibly points out to lower anxiety rate in this species. In the study of Birke et al.'s (1985) they assessed use of a burrow by acomys and mice (Birke, D'Udine, and Albonetti, 1985). Mice showed to spend more time in the closed environment of burrow, compared to acomys. Similarly in our work mice spent more time than acomys in a closed and darker environment of closed arms. At the same time, mice spent more time in the central part of the arena. In Birke et al.'s work (1985) mice had shown higher numbers of peeping and stretching — a behavior marked by authors as "undecisive", that could be similar to preferring to stay at the central part of the arena, but not enter the open arms. At the same time, most assessed parameters of the light/dark box test were similar for both acomys and mouse. Note that the light/dark box test is also used for assessing the preference of nocturnal rodents for dark areas (Crawley and Goodwin, 1980), therefore, our results indicated that both species may have a similar preference for dark environments. However, in light/dark box test acomys had lower latency time to come out of the dark part of the arena into the light. That, again, is consistent with behavior observations in previous work. Despite the fact that the conditions in our work and works that were done previously differ significantly, similar behavioral traits can be noted.

A dependency of the acomys's locomotor activity on the illumination was documented previously; longer trips were observed in an illuminated or open arena, in contrast to the dark arena, authors also noted that in illuminated arena acomys moved mainly along the walls (Eilam, 2004). However, in that paper, the arena was illuminated by two 300 W light bulbs. It led to a much higher illumination level compared to our conditions. Possibly, it is the reason for the reduced moving activity of our acomyses in the open field test. The same fact can possibly be the result of the discrepancy between our and data presented in Ratnayake et al., 2014. They reported the walking distance in the central part of the open field arena accounted for 10 % of the total walking distance (Ratnayake et al., 2014), which is lower than the present findings (ranging from 24 % to 41 %); the lighting in that study was low (2.8 lux).

It is important to note that some studies suggest that acomyses are more closely related to gerbils than to mice (Chevret et al., 1993; Agulnik and Silver, 1996; Gustavsen, Kvicerova, Dickinson, and Heller, 2009). In this context, it is interesting that the percentage of time spent in the open arms of the elevated plus-maze in gerbils (approximately 32 %) (Varty et al., 2002) is similar to the present findings in acomyses.

Conclusions

Some differences in exploratory behavior between *Acomys cahirinus* and *Mus musculus* were reported, possibly indicating higher exploratory behavior and lower anxiety of acomyses compared to mice.

References

Agulnik, S. I. and Silver, L. M. 1996. The Cairo spiny mouse Acomys cahirinus shows a strong affinity to the Mongolian gerbil Meriones unguiculatus. *Molecular Biology and Evolution* 13(1):3–6. https://doi.org/10.1093/oxfordjournals.molbev.a025567

Balkaya, M. G., Trueman, R. C., Boltze, J., Corbett, D., and Jolkkonen, J. 2018. Behavioral outcome measures to improve experimental stroke research. *Behavioural Brain Research* 352:161–171. https://doi.org/10.1016/j.bbr.2017.07.039

Birke, L. I. and Sadler, D. 1986. Patterns of exploratory behavior in the spiny mouse, *Acomys cahirinus*. *Behavioral and Neural Biology* 45(1):88–106. https://doi.org/10.1016/s0163-1047(86)80009-7

Birke, L. I., D'Udine, B., and Albonetti, M. E. 1985. Exploratory behavior of two species of murid rodents, *Acomys cahirinus* and *Mus musculus*: A comparative study. *Behavioral and Neural Biology* 43(2):143–161. https://doi.org/10.1016/s0163-1047(85)91337-8

Callaghan, R. M., Yang, H., Moloney, R. D., and Waeber, C. 2025. Behavioural assessment of neuropsychiatric outcomes in rodent stroke models. *Journal of Cerebral Blood Flow and Metabolism*. https://doi.org/10.1177/0271678X251317369

Chevret, P., Denys, C., Jaeger, J. J., Michaux, J., and Catzeflis, F. M. 1993. Molecular evidence that the spiny mouse (Acomys) is more closely related to gerbils (Gerbillinae) than to true mice (Murinae). *Proceedings of the National Academy of Sciences of the United States*

PHYSIOLOGY

of America 90(8):3433–3436. https://doi.org/10.1073/pnas.90.8.3433

Clemens, L. E., Jansson, E. K., Portal, E., Riess, O., and Nguyen, H. P. 2014. A behavioral comparison of the common laboratory rat strains Lister Hooded, Lewis, Fischer 344 and Wistar in an automated homecage system. *Genes, Brain and Behavior* 13(3):305–321. https://doi.org/10.1111/gbb.12093

Cohen, R., Kronfeld-Schor, N., Ramanathan, C., Baumgras, A., and Smale, L. 2010. The substructure of the suprachiasmatic nucleus: Similarities between nocturnal and diurnal spiny mice. *Brain, Behavior and Evolution* 75(1):9–22. https://doi.org/10.1159/000282172

Crawley, J. and Goodwin, F. K. 1980. Preliminary report of a simple animal behavior model for the anxiolytic effects of benzodiazepines. *Pharmacology, Biochemistry, and Behavior* 13(2):167–170. https://doi.org/10.1016/0091-3057(80)90067-2

Crusio, W. E., Schwegler, H., Brust, I., and Van Abeelen, J. H. 1989. Genetic selection for novelty-induced rearing behavior in mice produces changes in hippocampal mossy fiber distributions. *Journal of Neurogenetics* 5(1):87–93. https://doi.org/10.3109/01677068909167267

Cryan, J. F. and Sweeney, F. F. 2011. The age of anxiety: Role of animal models of anxiolytic action in drug discovery. *British Journal of Pharmacology* 164(4):1129–1161. https://doi.org/10.1111/j.1476-5381.2011.01362.x

Dmitrieva, N. I., Danilenkova, L. V., Gozzo, S., and D'Udine, B. 1987. Comparative study of the brain of developed and underdeveloped neonatal rodents of the *Muridae* family. *Zhurnal evoliutsionnoi biokhimii i fiziologii* 23(1):127–132. (In Russian)

D'Udine, B. and Gozzo, S. 1983. Archicortex and neocortex in the precocial murid *Acomys cahirinus*. A comparison with two altricial species: *Mus musculus* and *Rattus norvegicus*. *The International Journal of Neuroscience* 20(3–4):255–263. https://doi.org/10.3109/00207458308986579

D'Udine, B., Dimitriev, Y., and Birke, L. I. A. 1987. Patterns of exploration of novel objects in two species of murid rodent, *mus musculus l.* and *acomys cahirinus* (desmarest). *Monitore Zoologico Italiano — Italian Journal of Zoology* 21(1):77–86. https://doi.org/10.1080/00269786.1987.10736521

Eilam, D. 2004. Locomotor activity in common spiny mice (*Acomys cahirinuse*): The effect of light and environmental complexity. BMC Ecology 4(1):1–10. https://doi.org/10.1186/1472-6785-4-16

Frynta, D., Tomanová, L., Holubová, K., Vobrubová, B., Štolhoferová, I., and Rudolfová, V. 2024. Structural consistency of exploratory behaviour of sub-adult and adult spiny mice (*Acomys cahirinus*) in seven different tests. *Behavioural Processes* 216:105003. https://doi.org/10.1016/j.beproc.2024.105003

Gould, T. D., Dao, D. T., and Kovacsics, C. E. 2009. The open field test; pp. 1–20 in: Gould, T. D. (ed.), *Mood and Anxiety Related Phenotypes in Mice*. Humana Press, Totowa, NJ.

Gozzo, S., Dimitrieva, N., Iacopino, C., and D'Udine, B. 1985. A comparative study of mossy fiber distribution in the brain of the precocial Acomys cahirinus and of the altricial Rattus norvegicus: Neuroanatomical bases and behavioral correlates. *The International Journal of Neuroscience* 28(3–4):163–172. https://doi.org/10.3109/00207458508985387

Greenberg, G. 1986. Depth perception in Mongolian gerbils (*Meriones unguiculatus*) and spiny mice (*Acomys russatus* and *A. cahirinus*). *Journal of Comparative Psychology* 100(1):81–84.

Gustavsen, C. R., Kvicerova, J., Dickinson, H., and Heller, R. S. 2009. Acomys, the closest relatives to Gerbils, do express Pdx-1 protein and have similar islet morphology to Gerbils. *Islets* 1(3):191–197. https://doi.org/10.4161/isl.1.3.9557

Günter, A., Belhadj, S., Seeliger, M. W., and Mühlfriedel, R. 2024. The Mongolian gerbil as an advanced model to study cone system physiology. *Frontiers in Cellular Neuroscience* 18:1339282. https://doi.org/10.3389/fncel.2024.1339282

Hagbi, Z. and Eilam, D. 2022. On heights and plains: How rodents from different habitats cope with three-dimensional environments? *PLOS One* 17(3):e0265176. https://doi.org/10.1371/journal.pone.0265176

Hall, F. S., Huang, S., Fong, G. W., Sundstrom, J. M., and Pert, A. 2000. Differential basis of strain and rearing effects on open-field behavior in Fawn Hooded and Wistar rats. *Physiology and Behavior* 71(5):525–532. https://doi.org/10.1016/s0031-9384(00)00372-3

Henry, B. L., Minassian, A., Young, J. W., Paulus, M. P., Geyer, M. A., and Perry, W. 2010. Cross-species assessments of motor and exploratory behavior related to bipolar disorder. *Neuroscience and Biobehavioral Reviews* 34(8):1296–1306. https://doi.org/10.1016/j.neubiorev.2010.04.002

Johnson, A., Varberg, Z., Benhardus, J., Maahs, A., and Schrater, P. 2012. The hippocampus and exploration: Dynamically evolving behavior and neural representations. *Frontiers in Human Neuroscience* 6:216. https://doi.org/10.3389/fnhum.2012.00216

Keeley, R. J., Bye, C., Trow, J., and McDonald, R. J. 2015. Strain and sex differences in brain and behaviour of adult rats: Learning and memory, anxiety and volumetric estimates. *Behavioural Brain Research* 288:118–131. https://doi.org/10.1016/j.bbr.2014.10.039

Kraeuter, A.-K., Guest, P. C., and Sarnyai, Z. 2019. The elevated plus maze test for measuring anxiety-like behavior in rodents; pp. 69–74 in: Guest, P. C. (ed.), *Pre-Clinical Models*. Springer New York, New York.

Kulesskaya, N. and Voikar, V. 2014. Assessment of mouse anxiety-like behavior in the light-dark box and open-field arena: role of equipment and procedure. *Physiology and Behavior* 133:30–38. https://doi.org/10.1016/j.physbeh.2014.05.006

Lever, C., Burton, S., and O'Keefe, J. 2006. Rearing on hind legs, environmental novelty, and the hippocampal formation. *Reviews in Neurosciences* 17(1–2):111–133. https://doi.org/10.1515/revneuro.2006.17.1-2.111

Merkulyeva, N. 2024. Comparative review of the brain development in *Acomys cahirinus*. *Neuroscience and Biobehavioral Reviews* 167:105939. https://doi.org/10.1016/j.neubiorev.2024.105939

Morrow, A., Smale, L., Meek, P. D., and Lundrigan, B. 2024. Trade-offs in the sensory brain between diurnal and nocturnal rodents. *Brain, Behavior and Evolution* 99(3):123–143. https://doi.org/10.1159/000538090

Nauman, D. J. 1968. Open field behavior of the Mongolian gerbil. *Psychonomic Science* 10:163–164. https://doi.org/10.3758/BF03331461

Nöldner, M. and Schötz, K. 2007. Inhibition of lipopolysaccharid-induced sickness behavior by a dry extract from the roots of Pelargonium sidoides (EPs 7630) in mice. *Phytomedicine: International Journal of Phytotherapy and Phytopharmacology* 6:27–31. https://doi.org/10.1016/j.phymed.2006.11.013

Prut, L. and Belzung, C. 2003. The open field as a paradigm to measure the effects of drugs on anxiety-like behaviors:

A review. *European Journal of Pharmacology* 463(1–3):3–33. https://doi.org/10.1016/s0014-2999(03)01272-x

Rampon, C. and Tsien, J. Z. 2000. Genetic analysis of learning behavior-induced structural plasticity. *Hippocampus* 10(5):605–609. https://doi.org/10.1002/1098-1063(2000)10:5<605::AID-HIPO11>3.0.CO;2-3

Ratnayake, U., Quinn, T. A., Castillo-Melendez, M., Dickinson, H., and Walker, D. W. 2012. Behaviour and hippocampus-specific changes in spiny mouse neonates after treatment of the mother with the viral-mimetic Poly I:C at mid-pregnancy. *Brain, Behavior, and Immunity* 26(8):1288–1299. https://doi.org/10.1016/j.bbi.2012.08.011

Ratnayake, U., Quinn, T., Daruwalla, K., Dickinson, H., and Walker, D. W. 2014. Understanding the behavioural phenotype of the precocial spiny mouse. *Behavioural Brain Research* 275:62–71. https://doi.org/10.1016/j.bbr.2014.08.035

Rico, J. L., Hurtado-Parrado, C., Vásquez-Sepúlveda, J., Fonseca, J., and Cardona, Á. 2017. El tiempo en el área central del laberinto en cruz elevado correlaciona con medidas relacionadas a impulsividad durante una tarea operante. *Universitas Psychologica* 15(5). https://doi.org/10.11144/Javeriana.upsy15-5.tcae

Rogov, G. A., Toropova, K. A., Rogozhnikova, O. S., Oleinichenko, V. Y., and Ivashkina, O. I. 2025. Comparative analysis of the exploratory behavior of laboratory mice and rodents from natural populations in standard and enriched versions of the open field test. *Neuroscience and Behavioral Physiology* 55:1143–1154. https://doi.org/10.1007/s11055-025-01871-3

Shimada, T., Matsumoto, K., Osanai, M., Matsuda, H., Terasawa, K., and Watanabe, H. 1995. The modified light/dark transition test in mice: Evaluation of classic and putative anxiolytic and anxiogenic drugs. *General Pharmacology* 26(1):205–210. https://doi.org/10.1016/0306-3623(94)00148-g

Shkorbatova, P. Yu., Veshchitskii, A. A., Mikhalkin, A. A., Nikitina, N. I., Belyaev, A. V., and Merkulyeva, N. S. 2024. Breeding of the Cairo spiny mouse (*Acomys cahirinus*) in laboratory conditions. *Journal of Evolutionary Biochemistry and Physiology* 60(4):1347–1362. https://doi.org/10.1134/S0022093024040082

Sturman, O., Germain, P.-L., and Bohacek, J. 2018. Exploratory rearing: A context- and stress-sensitive behavior recorded in the open-field test. *Stress* 21(5):443–452. https://doi.org/10.1080/10253890.2018.1438405

Tomasso, A., Disela, V., Longaker, M. T., and Bartscherer, K. 2024. Marvels of spiny mouse regeneration: Cellular players and their interactions in restoring tissue architecture in mammals. *Current Opinion in Genetics and Development* 87:102228. https://doi.org/10.1016/j.gde.2024.102228

Tsao, C.-H., Wu, K.-Y., Su, N. C., Edwards, A., and Huang, G.-J. 2023. The influence of sex difference on behavior and adult hippocampal neurogenesis in C57BL/6 mice. *Scientific Reports* 13(1):17297. https://doi.org/10.1038/s41598-023-44360-8

Tucker, L. B. and McCabe, J. T. 2017. Behavior of male and female C57BL/6J mice Is more consistent with repeated trials in the elevated zero maze than in the elevated plus maze. *Frontiers in Behavioral Neuroscience* 11:13. https://doi.org/10.3389/fnbeh.2017.00013

Varty, G. B., Morgan, C. A., Cohen-Williams, M. E., Coffin, V. L., and Carey, G. J. 2002. The gerbil elevated plus-maze I: Behavioral characterization and pharmacological validation. *Neuropsychopharmacology* 27(3):357–370. https://doi.org/10.1016/S0893-133X(02)00312-3

Wilson, R. C., Vacek, T., Lanier, D. L., and Dewsbury, D. A. 1976. Open-field behavior in muroid rodents. *Behavioral Biology* 17:495–506. https://doi.org/10.1016/S0091-6773(76)90901-9

Zeng, Y.-F., Yang, K.-X., Cui, Y., Zhu, X.-N., Li, R., Zhang, H., Wu, D. C., Stevens, R. C., Hu, J., and Zhou, N. 2024. Conjunctive encoding of exploratory intentions and spatial information in the hippocampus. *Nature Communications* 15(1):3221. https://doi.org/10.1038/s41467-024-47570-4